

Mathematics A

Unit 1 • Chapter OL-T28 Classified

Cross-session • chapter-organised candidate

2 classified items • 9 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Unit 1 • Chapter OL-T28 • 2 items • 9 marks

Chapter	Items	Marks	Starts
01. OL-T28 Forming & Solving Equations	2	9	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Forming & Solving Equations

Unit 1 • 9 marks • 2 item(s)

November 2021 | 4MA1-1H Q5 | 5 marks

Area or dimension quadratic model

November 2021 | 4MA1-2H Q2 | 4 marks

Area or dimension quadratic model

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

- 5 Yasmin has some identical rectangular tiles.
Each tile is L cm by W cm.

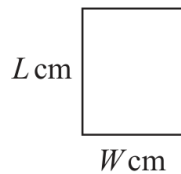


Diagram **NOT**
accurately drawn

Using 9 of her tiles, Yasmin makes rectangle $ABCD$, shown in the diagram below.

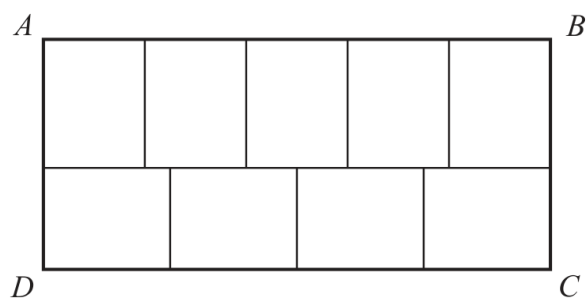


Diagram **NOT**
accurately drawn

The area of $ABCD$ is 1620 cm^2

Work out the value of L and the value of W .

$L = \dots\dots\dots$ $W = \dots\dots\dots$

Worked solution

November 2021 | Question 5 | 5 marks

Family: Form and solve a quadratic from data • Variant: Area or dimension quadratic model

Method / working

1. The top row has 5 tile widths and the bottom row has 4 tile lengths, so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $5W=4L$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $L=\frac{5}{4}W$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. The rectangle has width $(5W)$ and height $(L+W)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $5W(L+W)=1620$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. Substitute $(L=\frac{5}{4}W)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $5W\left(\frac{5}{4}W+W\right)=1620$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $5W\left(\frac{9}{4}W\right)=1620$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $\frac{45}{4}W^2=1620$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $W^2=144$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. $W=12$, $L=\frac{5}{4}(12)=15$

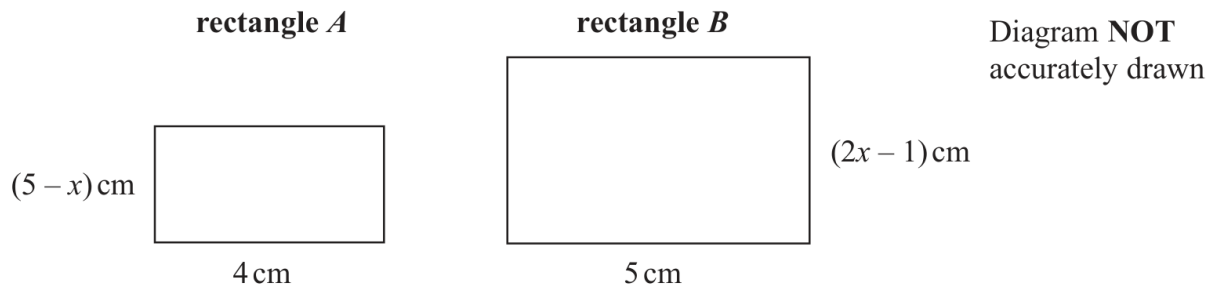
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. Answer: $(L=15 \text{ cm})$, $(W=12 \text{ cm})$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $L=15 \text{ cm}$, $W=12 \text{ cm}$.

- 2 Here are two rectangles, rectangle A and rectangle B .



The area of rectangle B is twice the area of rectangle A .

Work out the value of x .

Show your working clearly.

$x = \dots\dots\dots$

Worked solution

November 2021 | Question 2 | 4 marks

Family: Form and solve a quadratic from data • Variant: Area or dimension quadratic model

Method / working

1. Area of rectangle A:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $4(5-x)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. Area of rectangle B:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $5(2x-1)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. Rectangle B has twice the area of rectangle A:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $5(2x-1)=2 \cdot 4(5-x)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $10x-5=40-8x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $18x=45$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $x=2.5$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. Answer: $\{x=2.5\}$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $x=2.5$.